

Space and time

Tuesday, September 15, 2026

9:51 AM

Math

$$\vec{a} \cdot \vec{b} = ab \cos \theta$$

Two-argument arctangent (atan2)

Angle normalization

OK to reduce mod 2π or 360 degrees if it's always going to be used in a trig function.

Otherwise "secular terms"

- Do not normalize
- Use radians

Coordinate systems

1. Center: location of the origin
 - a. Center of the earth/gravitation (orbit center)
 - b. Observation site on surface
 - c. A satellite
2. Orientation: principal (first) axis, polar (third)
 - a. For earth-centered, polar axis is the north pole axis
 - b. For orbit-centered (perifocal) : angular momentum
 - c. Satellite-centered: angular momentum
3. Type: Cartesian, polar
 - a. Cylindrical
 - b. Spherical
 - c. Distance

Spherical to Cartesian (Matlab: `sph2cart`, AstroPy `spherical_to_cartesian`)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r \cos \phi \cos \theta \\ r \cos \phi \sin \theta \\ r \sin \phi \end{bmatrix},$$

Cartesian to spherical (Matlab: `cart2sph`, AstroPy `cartesian_to_spherical`)

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} \\ \theta &= \arctan(y, x) \\ \phi &= \arctan(z, \sqrt{x^2 + y^2}). \end{aligned}$$

Fig 1.18

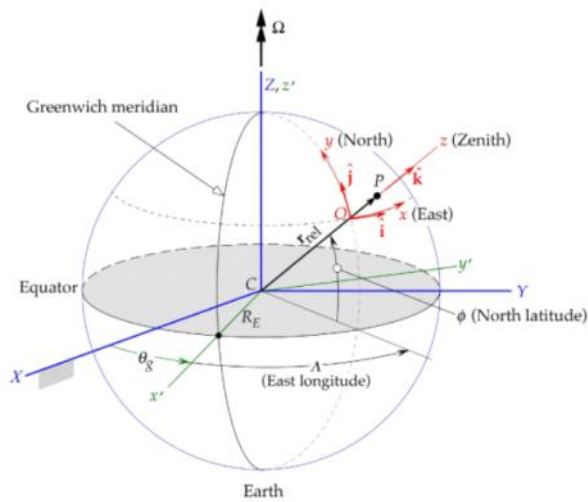


FIG. 1.18

Earth-centered inertial frame (XYZ) ; earth-centered noninertial $x'y'z'$ frame embedded in and rotating with the

Two earth-centered reference frames

- 1) Inertial: principal axis is fixed to the vernal equinox
- 2) Rotating (body fixed): principal axis points to the prime meridian

Celestial sphere

Earth sphere

Curtis sec. 4.2

Geocentric equatorial (IJK)

I = principal axis, points to vernal equinox

K = polar axis



Polar angles

- Right ascension α
- Declination δ

Equinox, or ascending node

Precession of the equinoxes

Nutation

Polar motion

Earth-centered earth-fixed ECEF

Longitude and latitude

Solar and sidereal time

Solar day: noon to noon 24 hours exactly,

Sidereal day:

Polar angles

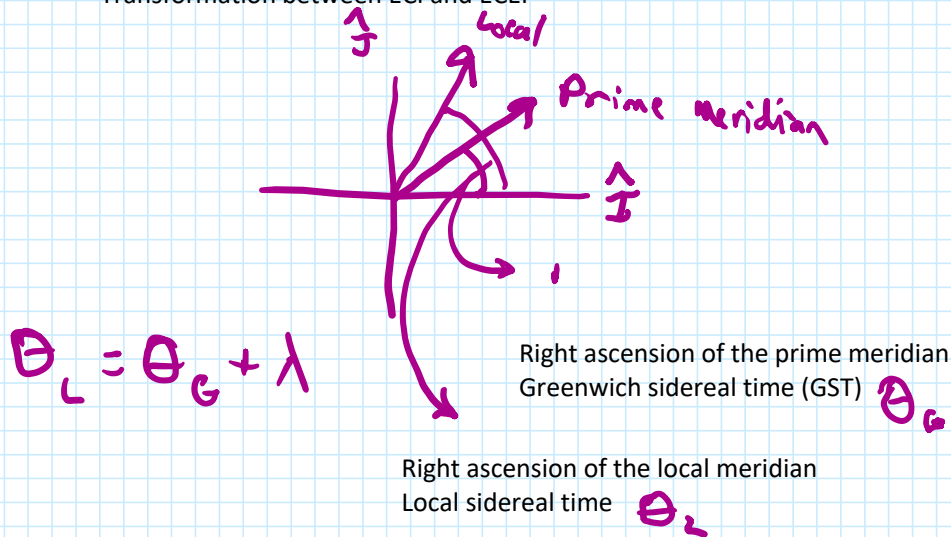
- Longitude λ
- Latitude ϕ

86164.090524 second
23:56:04s = sidereal day



23:56:04s = sidereal day

Transformation between ECI and ECEF



Transform by rotating between the two frames.

Lecture 6
9/17/2026

Time

- Origin and unit
- Epoch time of orbital motion

Julian day
Modified Julian day

Universal time - UT
UTC - Coordinated universal time
UT1 - Astronomical time
UTC-UT1 was previously limited to 0.9 s, engendering the need for leap seconds
Leap seconds are on their way out, to phased out officially in ~2035

TAI - international atomic time
GPS = TAI - 19s
TT terrestrial time TAI + 32.184s
TAI-UTC = 37s

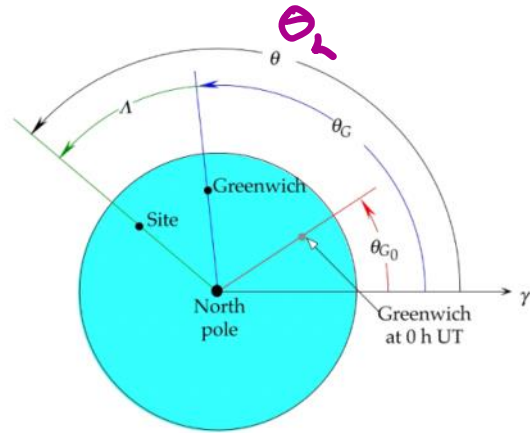


FIG. 5.7

Schematic of the relationship among θ_G , θ_L , Λ , and θ .

$$\Theta_L = \Theta_G + \Lambda$$

In one sidereal day, sidereal time changes by 360 degrees or 2 pi radians
That takes 86164.09s

Precision reference frames and systems

Celestial motion of the spin axis

- Precession 25,800 years
- Nutation, shorter period oscillation (up to 18.6 years)

Terrestrial motion, sub-daily

- Polar motion ~10m
- Tidal & relativistic effects: ocean/earth tides cause sub-daily variations, earth rotation speed (UT1)

Inertial frames

- GCRF - geocentric celestial reference frame
- EME2000 (formerly J2000) earth mean equator & equinox of J2000
- TEME - true earth, mean equinox

Epoch time of the inertial frame = J2000 = January 1, 2000 12:00 TT

EME2000 and GCRF have ~20 mas difference, we will not worry about the difference

Mean equator, mean equinox - EME2000, GCRF

Special reference from TEME (true equator, mean equinox) - for mean elements (explained later) considered much less accurate

"Mean" - includes only precession (25800 year period), and the nutation is averaged

"True" - the actual, instantaneous direction including both precession and nutation

"Mean of date", "true of date"

- Of date - coordinates and reference planes - instantaneous moment of

observation

- Of epoch - fixed at specific fundamental reference time, such J2000.0

GMST - Greenwich mean sidereal time (averaged equinox)

GAST - Greenwich apparent sidereal time (true equinox)

Connected by the "equation of the equinoxes"

GAST = GMST + nutation in the right ascension

GST = GAST

Solar system ephemerides

Where are all the bodies of the solar system (and what are their properties)?

JPL development ephemeris series (DE)

DE440/441 (2021)

De440 is accurate for the 1550-2650

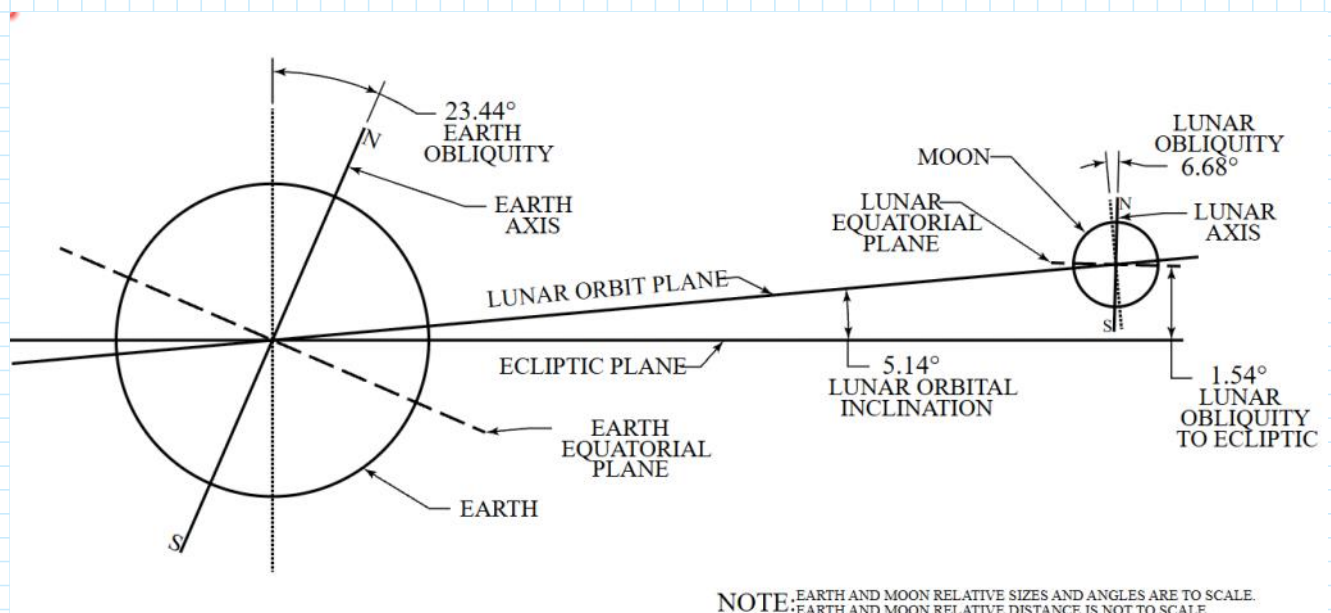
DE441 longer period -13,200 to + 17191 years

Horizons - locations of all the bodies in the DE, plus asteroids, comets, planetary satellites, Lagrange points

DE440 aligned to ICRF - international celestial reference frame, from which the GCRF is derived

SPICE toolkit - astronomical data available locally, e.g. de440.bsp

Lunar coordinate frames



Non-rotating principal axis: defined by the cross produce of the earth's mean rotational pole of J2000 with the moon's rotational north pole.

Rotating frame

- PA - principal axis - based on moments of inertia
- ME - mean earth (older)

Satellite centered coordinate systems

- Center: particular satellite of interest
- Type: usually Cartesian
- Orientation: polar axis is $W = h$, perpendicular to the orbital plane
 - RSW principal axis aligns with radial vector
 - NTW *second* axis aligns with velocity vector

Lecture 7
9/22/2026

Homework 2 is due September 30

RSW = local vertical local horizontal "LVLH"

No used here

Radial-in-track-crosstrack "RIC"



Along-track

Radial-transverse-normal "RTN"

RSW

R = radial direction

W = angular momentum direction

$S = W \times R$

NTW

T = velocity vector direction

W = angular momentum direction

$N = T \times W$